

VINCE PAUL P. JUGUILON

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SUMMARY

Multidisciplinary professional and Physics PhD candidate with a unique expertise at the intersection of **Experimental Physics**, **Data Science**, and **Intellectual Property**.

EDUCATION

Doctor of Philosophy in Physics University of the Philippines Diliman	<i>Ongoing</i>
<i>Research Topic: Terahertz Spintronics and Orbitronics (Coursework in progress)</i>	
Master of Science in Physics University of the Philippines Diliman	<i>June 2023</i>
<i>Thesis Title: Ultrafast carrier dynamics in InAs/GaAs self-assembled quantum dots investigated via optical pump – terahertz probe spectroscopy</i>	
Bachelor of Science in Applied Physics University of the Philippines Diliman	<i>February 2021</i>
<i>Thesis Title: Design of an absolute spectral emission power distribution measurement system in the UV-VIS-NIR region using an integrating sphere head and spectrometer</i>	
Secondary Education Philippine Science High School	<i>March 2013</i>

PROFESSIONAL EXPERIENCE

Patent Analyst Parola Analytics	<i>June 2024 - Current</i>
Project Leadership: Led and contributed to dozens of patent research projects, performing data-backed patentability and freedom-to-operate (FTO) analyses across diverse technical fields such as electronic engineering, mechanical engineering, IT, and materials science. Data Acquisition & ETL: Engineered automated data pipelines to extract and aggregate large-scale patent datasets from databases such as Orbit Intelligence, Google Patents (web scraping), US Patent and Trademark Office – Open Data Portal (ODP) API, European Patent Office – Open Patent Services (OPS) API, and public datasets on BigQuery. Data Wrangling & Entity Resolution: Performed extensive data cleaning and normalization on heterogeneous assignee and inventor datasets. Implemented Fuzzy Matching algorithms and String Distance metrics (such as Levenshtein) using Python (Pandas) to resolve duplicates and ensure high accuracy in entity resolution for downstream modeling. NLP & Advanced Analytics: Applied Natural Language Processing (NLP) techniques to unstructured patent abstracts, utilizing Transformer-based models (BERT) for Semantic Search and Topic Modeling. Conducted Time Series Analysis and Predictive Trend Modeling to identify emerging technology white spaces and competitive shifts. Business Intelligence & Visualization: Developed interactive dashboards in Power BI and Looker Studio to present complex landscape analyses to stakeholders, directly influencing client IP strategies. Workflow Automation & Data Loading: Engineered custom Google Apps Script solutions and REST API integrations (including EPO OPS and USPTO PatentsView) to automate high-volume data ingestion and report formatting, significantly reducing the teams' manual processing cycles. Data Storytelling & Dissemination: Authored SEO-optimized technical reports and blog posts to communicate complex patent landscape findings in order to translate high-dimensional data into actionable insights for diverse stakeholders. Hardware-to-Claims Mapping: Managed complex Evidence-of-Use (EOU) projects by applying scientific characterization techniques (such as optical microscopy, SEM, and EDX) to map physical hardware components to specific technical limitations within patent claims. Stakeholder Management & Technical Consulting: Presented data-driven findings and Evidence-of-Use (EOU) results to international clients and law firms.	
University Instructor University of the Philippines Diliman	<i>September 2021 – January 2024</i>
Technical Instruction: Delivered lecture and laboratory courses for engineering and physics majors, specializing in Classical Mechanics, Electromagnetism, Optics, Electronics, and Introductory Programming. Curriculum Engineering & Leadership: Served as Course Group Leader for an elementary physics laboratory, where I designed experimental curricula and authored technical manuals to standardize instruction and data collection methods. Project Management & Logistics: Coordinated equipment logistics and laboratory workflows to ensure the smooth delivery of technical experiments for large student cohorts. Technical Mentorship & Research Review: Acted as a Thesis Defense Panelist for undergraduate research, providing critical feedback on experimental rigor, data analysis methodology, and the quality of technical documentation. Cross-functional Team Coordination: Collaborated with a team of faculty and instructors throughout the semester to ensure consistency in syllabus delivery across multiple course sections.	

- **Computational Modeling & Numerical Analysis:** Developed **Python-based computational models** to simulate carrier dynamics by solving complex **Partial Differential Equations (PDEs)** and performed **Model Validation** by statistically comparing computational simulations with experimental data to ensure predictive accuracy.
- **System Engineering & Data Acquisition:** Assisted in the execution of a **DOST-funded research project** focused on the development of fast-scan **Terahertz (THz) Spectroscopy**. Contributed to **experimental design, project scoping**, and the **procurement** of high-precision laboratory instrumentation.
- **Experimental Research & Signal Processing:** Conducted research on **THz spectroscopy**, managing and maintaining high-sensitivity optical setups and laser systems to ensure the integrity of experimental datasets.
- **International Collaboration & Analytics:** Collaborated with **international research institutions** on joint experiments and data analysis, contributing to the publication of results across global, cross-functional teams.
- **Technical Communication:** Presented research findings at **international conferences** and published results in **peer-reviewed journals** and **conference proceedings**.

CERTIFICATIONS

- [Datacamp Data Scientist Associate](#)
- [Google Business Intelligence Professional Certificate](#)
- [Datacamp Data Analyst Associate](#)
- [Google Data Analytics Professional Certificate](#)
- [Datacamp SQL Associate](#)

SKILLS

Data Science & Engineering

- **Programming:** Python (Pandas, Scikit-learn, SciPy, TensorFlow, PyTorch), SQL (Advanced Queries & Window Functions), MATLAB, Google Apps Script
- **Machine Learning:** Natural Language Processing (BERT/Transformers), Time Series Analysis, Predictive Modeling, Entity Resolution (Fuzzy Matching), Statistical Modeling
- **Visualization & BI:** Power BI, Looker Studio, Tableau, Seaborn, Matplotlib, OriginLab
- **Pipeline Development:** End-to-end ETL Pipeline development, REST API integration (USPTO PatentsView, EPO OPS), Web Scraping (Google Patents)
- **Cloud & Databases:** Google BigQuery (Google Cloud Platform), Advanced SQL & Database Management, Data Normalization and Cleaning
- **Workflow & Tools:** Version Control (Git/GitHub), Virtual Environments (Conda), Technical Documentation

Patent Research & Analysis

- **IP Strategy:** Patentability, Freedom-to-Operate (FTO), and Evidence-of-Use (EOU) Analyses
- **Tools:** Orbit Intelligence, Google Patents, WIPO Patentscope, USPTO Open Data Portal, Espacenet.

Experimental Physics & Research

- **Characterization:** Terahertz Time-Domain Spectroscopy, Ultrafast Optics, Sputtering and Thin Film Deposition, Semiconductor Growth and Fabrication
- **Hardware:** Arduino Platform, National Instruments LabView, Remote Sensing

TRAININGS, SEMINARS, AND CONFERENCES

- DL-101 General Course on Intellectual Property | World Intellectual Property Office | December 2025
- Basics of Quantum Information | IBM Quantum | November 2025
- PatentX: Patent Law and Global Public Health | World Intellectual Property Office and Harvard Law School | September 2024 to December 2024
- The 8th Academic Conference on Natural Science for Young Scientists, Master and PhD Students from ASEAN Countries (CASEAN) | Presenter | Vinh City, Vietnam | 2023
- International Collaborative Research Program | Invited Researcher (ICRP) | Fukui, Japan | 2023
- 3rd Philippines-Japan Terahertz Workshop (PJTW) | Speaker | Philippines | 2023
- 8th Southeast Asia Collaborative Symposium on Energy Materials | Presenter | Tsukuba, Japan | 2022
- 12th International Conference on Photonics and Applications (ICPA) | Presenter | Con Dao, Vietnam | 2022
- Fundamentals of University Teaching | UP Diliman – Office for the Advancement of Teaching | 2021